

User Insights and Interactions in Clustering

- ❑ Visual insights: One picture is worth a thousand words
 - ❑ Human eyes: High-speed processor linking with a rich knowledge-base
 - ❑ A human can provide intuitive insights; HD-eye: visualizing HD clusters
- ❑ Semi-supervised insights: Passing user's insights or intention to system
 - ❑ User-seeding: A user provides a number of labeled examples, approximately representing categories of interest
- ❑ Multi-view and ensemble-based insights
 - ❑ Multi-view clustering: Multiple clusterings represent different perspectives
 - ❑ Multiple clustering results can be ensembled to provide a more robust solution
- ❑ Validation-based insights: Evaluation of the quality of clusters generated
 - ❑ May use case studies, specific measures, or pre-existing labels